

A PROJECT TITLE

ON

**Forest Fire Risk Prediction Using Random Forest and XGBoost**

*project report submitted in partial fulfilment of the requirements for the Course*

**MACHINE LEARNING  
BACHELOR OF TECHNOLOGY**

in

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by

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## 1. Abstract:

Forest fires are among the most destructive natural disasters, causing severe environmental, economic, and social impacts. They lead to the loss of biodiversity, destruction of vegetation, degradation of air quality, and significant threats to human life and property. Climate change, rising temperatures, prolonged droughts, and increasing human activities have further increased the frequency and intensity of forest fires worldwide. Therefore, developing an accurate and reliable fire risk prediction system has become essential for effective forest management and disaster prevention. This project presents a **Forest Fire Risk Prediction System** using two advanced ensemble machine learning algorithms: **Random Forest** and **XGBoost (Extreme Gradient Boosting)**. These algorithms are selected because of their high predictive accuracy, ability to handle large datasets, resistance to overfitting, and capability to model complex relationships between environmental variables. The system analyzes historical forest fire data consisting of factors such as temperature, relative humidity, wind speed, rainfall, soil moisture, drought index, vegetation condition, and other meteorological parameters that significantly influence fire occurrence.

The dataset undergoes extensive preprocessing, including handling missing values, removing duplicate records, encoding categorical variables, feature scaling where necessary, and feature selection to improve prediction quality. Exploratory Data Analysis (EDA) is performed to understand data distribution, identify trends, and determine the most influential factors contributing to forest fire outbreaks. The processed data is then divided into training and testing sets for model development and evaluation.

## 2. INTRODUCTION:

Forest fires are one of the most devastating natural disasters, affecting millions of hectares of land every year. They cause significant damage to forests, wildlife habitats, agricultural land, and nearby human settlements. In addition to environmental destruction, forest fires release large amounts of carbon dioxide and other harmful gases into the atmosphere, contributing to global warming and climate change. The increasing frequency and severity of forest fires due to rising temperatures, prolonged droughts, and human activities have made early fire prediction an important area of research.

Traditionally, forest fire monitoring has relied on manual observation, satellite surveillance, and weather forecasting. Although these methods are useful, they often face limitations such as delayed detection, high operational costs, and reduced accuracy under changing environmental conditions. As the volume of environmental and meteorological data continues to grow, machine learning has emerged as a powerful tool for predicting forest fire risk by identifying hidden patterns and relationships within historical data.

Machine learning algorithms can analyze multiple environmental factors simultaneously, including temperature, humidity, rainfall, wind speed, soil moisture, vegetation conditions, and drought indices. By learning from historical fire incidents, these algorithms can estimate the probability of future fire occurrences with high accuracy. Such predictions enable forest authorities to take preventive measures, allocate firefighting resources efficiently, and reduce the overall impact of forest fires.

### 3. SOFTWARE REQUIREMENTS:

The following software components are required to develop and implement the Forest Fire Risk Prediction Using Random Forest and XGBoost project.

| Software                   | Purpose                                          |
|----------------------------|--------------------------------------------------|
| Operating System           | Windows 10/11, Linux, or macOS                   |
| Programming Language       | Python 3.10 or above                             |
| IDE                        | Jupyter Notebook, Visual Studio Code, or PyCharm |
| Machine Learning Library   | Scikit-learn                                     |
| Gradient Boosting Library  | XGBoost                                          |
| Data Manipulation          | Pandas, NumPy                                    |
| Data Visualization         | Matplotlib, Seaborn                              |
| Model Evaluation           | Scikit-learn Metrics                             |
| Dataset Format             | CSV (Comma Separated Values)                     |
| Version Control (Optional) | Git and GitHub                                   |

#### Required Python Libraries

- Pandas
- NumPy
- Scikit-learn
- XGBoost
- Matplotlib
- Seaborn
- Joblib (for saving trained models)

#### Software Installation

Install the required libraries using the following command:

```
pip install pandas numpy scikit-learn xgboost matplotlib seaborn joblib
```

#### Development Environment

- Python: 3.10+
- IDE: Visual Studio Code / Jupyter Notebook / PyCharm
- Framework: Scikit-learn

- **Machine Learning Models:** Random Forest, XGBoost
- **Dataset:** Forest Fire Dataset (CSV)
- **Visualization Tools:** Matplotlib and Seaborn

These software requirements provide the necessary environment for data preprocessing, model training, evaluation, visualization, and prediction of forest fire risks.



Signature of the Guide



Course Instructor